National University of Computer and Emerging Sciences, Lahore Campus

Course Name:	Object Oriented Programming	Course Code:	CS1004
Degree Program:	BS	Semester:	Summer 2023
Exam Duration:	120 Minutes	Total Marks:	40
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Sections:	BCS-9A/9C	No of Page(s):	8
Exam Type:	Midterm		

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Instruction/Notes: Attempt all questions. Do not use pencil or red ink. In case of confusion or ambiguity make a reasonable assumption. **Do not attach any extra sheet**. Use extra sheet for rough work only

Question 1: Write the Output of the following Code: (10 Marks) **(CLO-1)**

Part a)

```
#include<iostream>
using namespace std;
void function_A(int* p, int*& q)
{
	p = new int;
	*p = *q + 40;
	*q = *p + 10;
	delete p;
}
void function_B(int*& p, int* q)
{
	q = new int;
	*q = *p - 10;
	*p = *q - 1;
	function_A(p, q);
	delete q;
}
int main()
{
	int x = 100;
	int* ptr1 = &x;
	int* ptr2 = new int;
	*ptr2 = 200;
	cout << *ptr1 << " " << *ptr2 << endl;
	function_B(ptr1, ptr2);
	cout << *ptr1 << " " << *ptr2 << endl;
	function_A(ptr1, ptr2);
	cout << *ptr1 << " " << *ptr2;
	delete ptr2;
	return 0;
}
```

Rough:

Output:

100 200

~~1~~ 200

~~1~~ 250

4

3.5

Part b)

```
#include <iostream>
using namespace std;
class Book{
public:
    int id;
    Book();
    Book(int);
    ~Book(){
        cout << "Destructor id: " << id << endl;
    }
};
Book::Book(){
    cout << "Book created" << endl;
}
int main(){
    Book obj1;
    obj1.id = 5;
    cout << "Obj1 id is " << obj1.id << endl;
    int i = 1;
    while (i < 4){
        Book obj2;
        obj2.id = i;
        cout << "Obj2 id is " << obj2.id << endl;
        i++;
    }
    return 0;
}
```

~~Rough~~: Output:-

Book created
Obj1 id is 5
Book created
obj2 id is 1
Destructor id : 1
Book created
obj2 id is 2
Destructor id : 2
Book created
obj2 id is 3
Destructor id : 3
Destructor id : 5

5

Output:

2: (15 Marks)

asing Application (CLO-3)

hrasing is the process of rewording a text, often done for simplification or clarity.

u need to write a function that will take a character array as input and after creating a 2d string array it will return the double pointer. The prototype is given below:

string CreateDynamicArray(char ws[])**

Your function will receive a character array like this:

char ws[200]="abandon discontinue vacate#absent missing unavailable#cable wire#calculate compute determine measure#safety security refuge";

In the character array there are words and their synonyms. Entries are separated by a hash symbol (#).

a) Your first task is to create a dynamic 2D strings array that will store the words and their synonyms in different columns like the picture below.

0	ptr0	→	abandon	discontinue	vacate		
1	ptr1	→	absent	missing	unavailable		
2	ptr2	→	cable	wire			
3	ptr3	→	calculate	compute	determine	measure	
4	ptr4	→	safety	security	refuge		

b) Your second task is to take input of a word and then display its last synonym
For example if user will enter **absent** your program will show **unavailable**
Note: assume that user will always enter the first word of every column to replace like abandon, absent, cable etc

```
string** CreatDynamicArray (char ws[]){
  const char* delimeter= "#";
  const int maxWords = 100;
  const int maxsyn = 10;
  string** dynamicArray= new string*[maxwords];
  for (int i=0; i<maxwords; i++){
     dynamic array[i]= new string[maxsyn];
  }

  char * word = strtok(ws, delimeter);
  int wordindex = 0;
```

90 ^

```
while (word != NULL) {
    Char* syn = strtok(NULL, delimeter);
    int synindex = 0;
    while (syn != NULL) {
        dynamicArray[wordIndex][synindex] = syn
        syn = strtok[NULL, delimeter]
        wordindex ++;
    }
    return dynamicArray;
}
```

(6)

```
int main() {
    char ws[200] = "abandon discontinue vocate
                    # absent missing unavailable #
                    cable wire # calculate compute
                    determine measure # safety
                    security refuge";
    string ** result = creatDynamicArray(ws);

    for(int i=0; i<6; i++){
        for(int j=0; j<2; j++) {
            cout << result[i][j] << " ";
        }
        cout << endl;
    }
```

(2)

... element){

}
```
for(int i=0 ; i<6 ; i++){
delete[] result[i];
}
delete []result;

}
```

(CLO-2)

Question 3: (15 Marks)

Create a class **DynamicArray** that has the following data members and functions.

Data members
arraySize- Total size of array
totalElement- number of elements stored in an array. It will be initially zero then after each insertion it wi
incremented
st*- pointer to the array

Description about the functions:
InsertElementAtEnd- Insert Element at the end of the array i.e. if the array size is 10 and the array is filled till th
index then the element will be inserted at the 6th index
DeleteElementFromEnd- Delete Element from the end of the array i.e. if the array size is 10 and the array is filled
the 5th index then element on 5th index will be deleted
isEmpty – checks if there is no element then it will return true otherwise false
isFull- checks if the array is filled then it will return true otherwise false
PrintElements- Print all elements of the array
SearchElement- Search an element from the array and return its index. If it's not in the array then display an appropriate message.

Note:

- Make all the constructors, use proper access specifiers and create proper destructor.
- No need to write main function

11.75

```
class DynamicArray {
 private:
   int arraysize;
   int telement;
   int * st;
 Public:
  DynamicArray(int size){
       arraysize = size;
       telement = 0;
       st = new int [arraysize];
  }
  ~DynamicArray() {
       delete[] st;
  }
```

2

1

1

```
void insertElementAtEnd(int element){
    if(telement < arraysize){
        st[telement] = element;
        telement++;
    }
}
void deleteElementFromEnd(){
    if(telement > 0){
        telement--;
    }
bool isEmpty(){
    return telement==0;
}
bool isFill(){
    return telement == arraysize;
}
void printElement(){
    for(int i=0; i<telement; i++){
        cout << st[i] << " "; }
    cout << endl;
}
int searchElement(int element){
    for(int i=0 ; i<telement ; i++){
        if(st[i]= element){
            return i; }
    }
    return -1;
}
};
```

1
1
1
1
1
1.75